

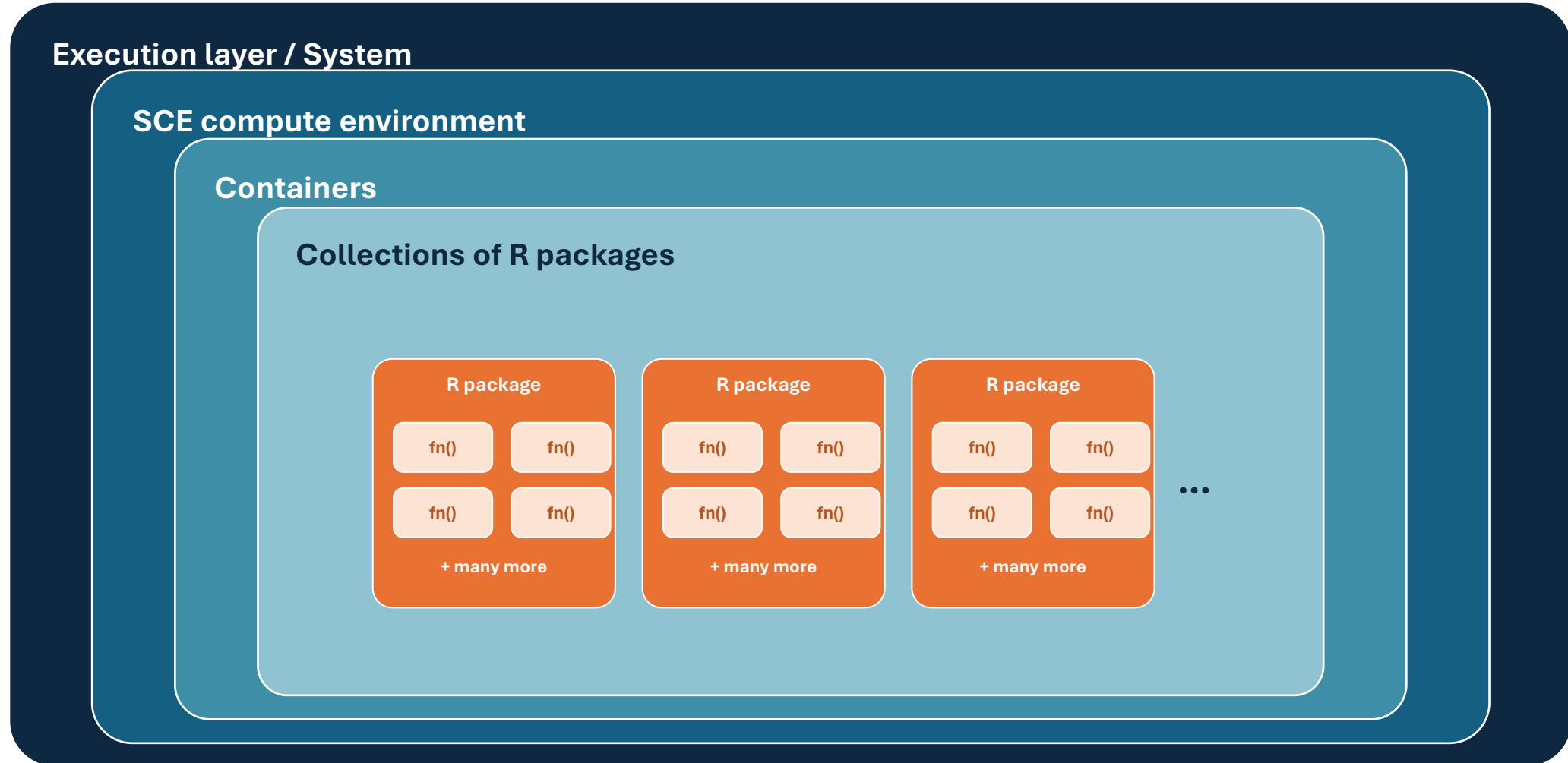
Attributes of “Production” environments

Before we get much further, let's try to agree on some terminology...

What do you mean “Production”?

- Regulatory-facing, or output from the system is involved in regulatory interactions.
- Anything that affects or records patient safety
- Anything that might affect dosing decisions
- Anything that leads to a decision being made
- GxP = **Good** <Clinical / Manufacturing / Lab / ...> **Practice**

What do you mean “Environment”?



What do you mean by “Validation”?

- “**Establishing documented evidence** which provides a high degree of assurance (*accuracy*) that a specific process consistently (*reproducibility*) produces a product meeting its predetermined specifications (*traceability*) and quality attributes.”
 - [*FDA Glossary of Computer System Terminology*](#)
- “The computer software used for data management and statistical analysis should be **reliable, and documentation of appropriate software testing procedures should be available.**”
 - [*ICH E9*](#)

What do you mean by “Validation”?

- “The sponsor may rely on qualification documentation provided by the vendor, if the qualification activities performed by the vendor have been assessed as adequate. However, the sponsor may also have to perform **additional qualification (and validation) activities based on a documented risk assessment.**” – EMA [*Notice to sponsors on validation and qualification of computerised systems used in clinical trials and the associated Q&A.*](#)

What do you mean by “Validation”?

- Additional Resources:
 - [R Consortium Validation Hub: Validation Overview](#)
 - [R Consortium Submissions Working Group](#)
 - [*R: Regulatory Compliance and Validation Issues A Guidance Document for the Use of R in Regulated Clinical Trial Environments*](#)
 - [*Posit – Manage Packages and Reproduce Environment.*](#)

***So... What do I mean “Attributes”
of a Production Environment?***

VERSION
CONTROLLED

VALIDATED

TESTED

REPRODUCIBLE

SECURE
COMPLIANT

CHANGE
CONTROL

ACCESS
CONTROL

TRACEABLE

GxP

STABLE
SUPPORTED

DOCUMENTED

MONITORED
GOVERNANCE

PROVENANCE

REPEATABLE

SNAPSHOT

BACKED UP

PEER REVIEWED

RISK BASED

SOPs

CONTAINERISED

IMMUTABLE

TRAINED USERS

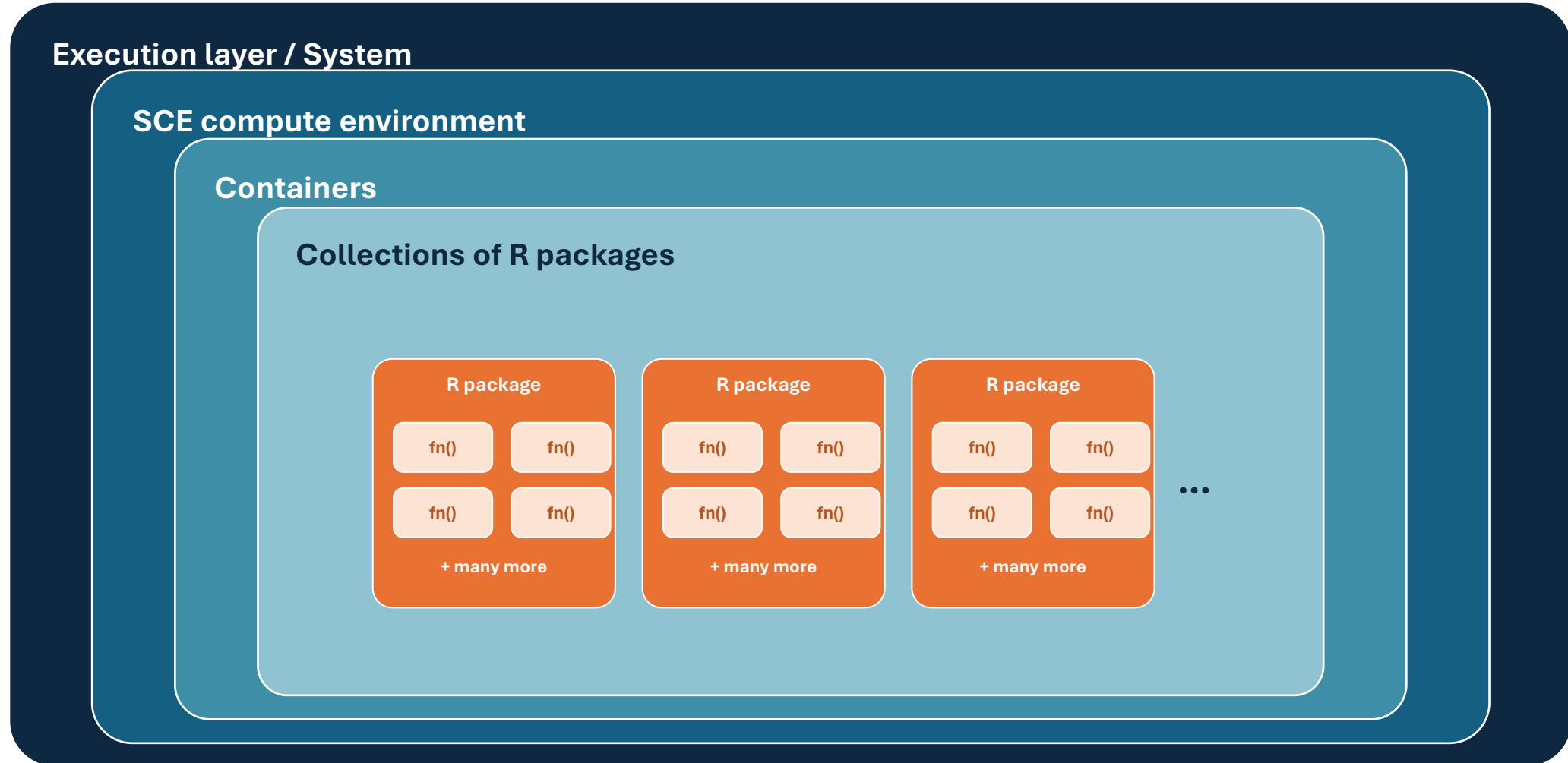
The aim of this whole day:

- Is there consensus which attributes are ***essential*** for Production Environments?
- Is there a way of fulfilling these regardless of your organisation size or budget?
- **When you return to your organisation, what conversations do you need to have to ensure you have at least the “minimum viable product” of a Production Environment meeting the essential attributes?**

Motivating use case

- Agency request to re-analyse data submitted to them 5 years ago, but looking at new subset of the data...
 - Step 1: re-analyse the original data to confirm previous result (reproducibility)
 - Step 2: Subset the data and re-run the analysis.
- If you know **NOW** that this is going to happen in 5 years (or 10 years!) then what would you want to put in place NOW to make sure you can address it in years to come?
- What attributes are essential? Which are desirable?

Bear in mind “Environment” = ALL OF THIS



OVER TO YOU...

In groups... **BRAINSTORMING**

- What are attributes of a Production Environment? **(5 mins)**
 - **THINK FAST!**
 - (We'll filter and refine later...)
 - Write down on sticky notes
- On the flip-charts, place your attribute under the “ESSENTIAL” or “Desirable” column. **(15 mins)**
 - **(Keep stickies ABOVE the horizontal line please!)**
 - **Discuss** within your group whether each is ESSENTIAL or Desirable
 - ESSENTIAL = **MUST** have it to fulfil regulatory expectations
 - Desirable = *everything else.*



In your groups...

- **Move** to a new flipchart and **REVIEW (10 mins)**
 - Review their ESSENTIAL and Desirable attributes
 - **Anything surprising?**
 - **Anything you missed?**
 - Re-categorize **THEIR** attributes by moving stickies to **BELOW** the horizontal line if you think any need to be reclassified.



Let's REVIEW

- **What attributes are deemed ESSENTIAL?**
 - Is there broad agreement?
 - Anything surprising?
 - It would be good to try to reach consensus on ESSENTIAL attributes.
- **Did anyone reclassify items?**
 - Why did you make that decision?



For discussion and consideration during the day:

- **ESSENTIAL** attributes
 - Is there a “Minimum Viable Implementation” for these?
- As we go through today, if you think of anything you want to add or reclassify, then please bring this up for discussion.

